

ECE 49595 SD I – Exercise 3 – Learn a skill

Fall 2026

Purpose of Assignment

- This assignment is designed to help you begin learning the technical skills your team will need to complete your project. The goal is to ensure that each team member dedicates focused time to hands-on exploration of their project's technical complexity.

General Instructions

- If you have questions about the assignment instructions or the scope / level of detail for what we are requesting, **ask questions early!**
- Do NOT submit long paragraphs with no visual distinction.
- DO use headers, section formatting, bullets, etc. to make your submission readable.
- Do NOT copy any assignment instructions into your submission.
- You are required to use a public GitHub repo to submit your code

Deliverable Scope:

- Deliverable and skill complexity will be measured as a part of the grading process.
- The expectation is that the completion of this assignment will take between 2-4 hours of work. Submitting a 30-line python file will not result in full marks on the assignment.
- Note that submitting previously completed from other classes for this assignment is plagiarism.
 - If previous work is related to the skill you want to focus on, you must produce a reasonable extension to the other project and clearly explain which portions are old vs. new.

GenAI Usage Statement

- Describe in 2-3 sentences how you used GenAI tools in the completion of the assignment, and identify specific portions of the submission that were authored by GenAI
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Identify and Justify a Relevant Skill

- Choose a technical skill and deliverable that will be directly applied within your project.
- Each team member must select one technical skill that is essential to the project.
 - Each team member should focus on a *different skill area* unless you provide a strong justification otherwise (e.g., if two members must both implement the same feature).

- Explain what technical skill you are developing, why this skill matters and which part of your project will rely on it.
- Explain why your chosen deliverable demonstrates your learning of the chosen skill.
- Examples of appropriate skills include, but are not limited to:
 - Writing PyTorch code to train a simple ML model.
 - Creating a basic iPhone app with multiple pages.
 - Hosting a website using React with navigation across sub-pages.
 - Using and managing an AWS database with CRUD operations.

Measurable Learning Objectives

- Describe what it means to successfully learn this skill.
- Define at least 3 specific, measurable learning objectives for yourself, that will demonstrate non-trivial skill acquisition.

Learn and Document

- Spend time learning your chosen skill.
- Provide links (or references for non-digital media) to the **most useful resources** (minimum 2) you found. Do not list everything you looked at—include only the ones you would actively recommend to other students learning the same skill.
- Explain why these resources were especially helpful.
 - Frame your recommendations as if you were advising a future student who is starting from the same point you were.
- Explain any limitations or gaps that you found in the resources.

Provide Evidence

- Include concrete evidence of your progress that demonstrates you learned the skill. Evidence should go beyond setup or installation.
- The required deliverables to submit in your Brightspace submission are:
 - A link to your GitHub repo with the project code and relevant documentation.
 - You are required to share that repo with the instructors so that they can access and review your code. The emails for their GitHub accounts are:
 - dubois6@purdue.edu
 - sfrance1@purdue.edu
 - sganguli@purdue.edu
 - Video from your development machine that shows you building and launching the code.
 - A description for each screenshot you provide.
- Examples of other appropriate evidence you may include:
 - Screenshot of model architecture code plus a training/validation accuracy plot.

- Unlisted YouTube video of you navigating through the app you coded.
- Link to a live website (or video demo) with working navigation between sub-pages.
- Screenshots showing successful requests to a REST API connected to AWS.

Reflect and Plan Next Steps

- Clearly identify the specific aspects of this skill that you have not yet mastered.
- Outline your step-by-step action plan to close these gaps.
 - Break it into milestones with specific deliverables. What timeline will you commit to completing each milestone?
 - Make sure your next steps are connected to your project. Do not state generic steps that would be relevant to anyone else learning the same skill.